Early rows of the reflected triangles, firing classes, and the five-state near misses

Research notes accompanying the note
Two necessary conditions on minimal-time solutions of the firing squad problem

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Notation and numbering of results refer to the note: the half-line C_∞, the line C_n, the reflected triangle R_n in the relative coordinates $\tau = t-(n-1)$, $\kappa = n-i$, so that $R_n = \{(\tau,\kappa) : 0 \le \kappa \le \tau \le n-1\}$, the input word

$$J_n(\kappa) = \big(\Phi_{n-3+\kappa}(2\kappa-2),\ \Phi_{n-2+\kappa}(2\kappa-1)\big), \qquad 0 \le \kappa \le n-1,$$

the depth rows $\Phi_t(j) = C_\infty(t, t+1-j)$ with the recurrence

$$\Phi_{t+1}(j) = \delta(\Phi_t(j), \Phi_t(j-1), \Phi_t(j-2)) \qquad (0 \le j \le t+1), \tag{1}$$

where $\Phi_t(-1) = \Phi_t(-2) = \mathsf{L}$ and $\Phi_t(t+1) = *$ (equation (3) of the note), and the partial map Ψ of the determinacy lemma (Lemma 3.1 of the note). We write $R_n(\tau,\kappa) := C_n(n-1+\tau, n-\kappa)$. All statements are proved below; the computer-assisted parts are simulations of explicitly given rule tables and use no SAT solver. They are reproduced by the programs `asymp.py` and `failall.py` in the directory `fssp5/research` of the repository.

1 Early rows

For $T \ge 0$ let $E(T) := \{(\tau,\kappa) : 0 \le \kappa \le \tau \le T\}$. If $n \ge T+2$, then $E(T) \subseteq R_n$ and no cell of $E(T)$ lies in the firing row $\tau = n-1$ of R_n.

Lemma 1 (early cells)**.** *Let δ be a rule, $T \ge 0$, and $n, n' \ge T+2$ with $J_n(\kappa) = J_{n'}(\kappa)$ for $0 \le \kappa \le T+1$. Then for every $(\tau,\kappa) \in E(T)$, $\Psi(J_n, n; \tau, \kappa)$ is defined if and only if $\Psi(J_{n'}, n'; \tau, \kappa)$ is defined, and then the two values are equal. In particular, the line of length n has a cell in state F at a relative time $\tau \le T$ if and only if the line of length n' has one at the same relative position.*

Proof. By the definition of Ψ in the proof of Lemma 3.1 of the note,

$$\Psi(J, m; \tau, \kappa) = \delta(x_{\kappa+1}, x_\kappa, x_{\kappa-1}),$$

where $x_{\kappa'}$ is the border $*$ if $\kappa' \in \{-1, m\}$, a component of $J(\kappa')$ if $\tau - 1 \in \{\kappa'-2, \kappa'-1\}$, and $\Psi(J, m; \tau-1, \kappa')$ otherwise (then $0 \le \kappa' \le \tau-1$). Unfolding this recursion from $(\tau,\kappa) \in E(T)$ visits only pairs (τ'', κ'') with $0 \le \kappa'' \le \tau'' \le T$ and reads the symbols $x_{\kappa'}$ with $\kappa' \le \kappa''+1 \le T+1$. The case $\kappa' = m$ would require $m \le T+1$, which is excluded for $m \in \{n, n'\}$; hence the left border is never read, and the input word is read only at positions $\kappa' \le T+1$, where J_n and $J_{n'}$ agree.

The two recursions are therefore identical step by step, including the points where an argument equal to F makes δ undefined. The last statement follows because the cells of the line of length n outside R_n coincide with cells of C_∞ (Lemma 2.2 of the note) and are never F before the line has fired, while the cells of R_n at the relative times $\tau \le T$ are exactly the cells of $E(T)$. □

Lemma 2 (periodicity certificate)**.** *Let δ be a rule whose half-line does not fire, $D \ge 0$, and $V_t := (\Phi_t(0), \dots, \Phi_t(D))$ for $t \ge D$. There is a map G_D, depending on δ and D only, with $V_{t+1} = G_D(V_t)$ for all $t \ge D$. Consequently, if $V_{t_0} = V_{t_0+p}$ for some $t_0 \ge D$ and $p \ge 1$, then $V_t = V_{t+p}$ for all $t \ge t_0$.*

Proof. For $t \ge D$ and $0 \le j \le D$ we have $j \le t$, so the three arguments in (1) are $\Phi_t(j)$, $\Phi_t(j-1)$, $\Phi_t(j-2)$ with indices at most D (and L for negative indices); none of them is the border symbol $\Phi_t(t+1) = *$. Hence $G_D(v)_j := \delta(v_j, v_{j-1}, v_{j-2})$, with $v_{-1} = v_{-2} = \mathsf{L}$, does it; it is well defined on the vectors that occur because the half-line does not fire. The second statement follows by induction on t. □

Corollary 3 (finitely many early-row problems)**.** *Let δ, $T \ge 0$, $D := 2T+1$, $t_0 \ge D$ and $p \ge 1$ satisfy $V_{t_0} = V_{t_0+p}$. Then $J_n(\kappa) = J_{n+p}(\kappa)$ for all $0 \le \kappa \le T+1$ and all $n \ge t_0 + 3$. Hence, for every $N_0 \ge T+2$, each line of length $n \ge N_0$ has the same cells at the relative times $\tau \le T$ (in the sense of Lemma 1) as one of the lines of the finitely many lengths $N_0 \le n' < \max(N_0, t_0+3) + p$.*

Proof. For $0 \le \kappa \le T+1$ the two components of $J_n(\kappa)$ are depth-row cells of depths $2\kappa-2, 2\kappa-1 \le 2T+1 = D$ at the times $n-3+\kappa$ and $n-2+\kappa$, which are at least $n-3 \ge t_0$ (for $\kappa = 0$ the lower component is L by convention, and the upper one is $C_\infty(n-2, n) = \mathsf{L}$ by Lemma 2.1 of the note). By Lemma 2 these cells are unchanged when n is replaced by $n+p$. If $n \ge \max(N_0, t_0+3) + p$, subtracting a suitable multiple of p gives n' in $[\max(N_0, t_0+3), \max(N_0, t_0+3)+p)$ with $J_{n'} = J_n$ on $[0, T+1]$, and Lemma 1 applies since $n, n' \ge N_0 \ge T+2$. □

Remark 4. For a minimal-time solution, every line of length $n \ge T+2$ has no F at the relative times $\tau \le T$. If the depth column V_t for $D = 2T+1$ is periodic from t_0 with period p, the early rows of R_n for $n \ge t_0 + 3$ depend only on $n \bmod p$: they are the first $T+1$ rows of finitely many *limit triangles*, which grow from the right border on the periodic background formed by the depth rows near the front. A minimal-time solution needs every limit triangle to stay free of F in all rows, just as the half-line itself never fires. The near misses of the note violate exactly this, as the next proposition shows.

Proposition 5. *None of the rules δ_{14}, δ_{14}^{b}, δ_{14}^{c}, δ_{14}^{d}, δ_{14}^{e} (files `results/delta14*.txt` of the repository) synchronizes a line of length $n \ge 15$ at time $2n-2$. Each of them therefore synchronizes exactly the lines of lengths $2, \dots, 14$ in minimal time.*

Proof (computer-assisted, by simulation). For each rule, Table 1 gives T and a repetition $V_{t_0} = V_{t_0+p}$ of the depth column for $D = 2T+4 \ge 2T+1$, found by simulating the half-line. Each rule's half-line uses only the transitions of its germ and does not fire on the computed cells, so Lemma 2 applies. By direct simulation:

(i) every line of length $15 \le n < T+3$ fails to synchronize at time $2n-2$ (early or incomplete firing);

(ii) for every n with $\max(15, T+3) \le n < t_0 + 3 + p$, the line of length n, or a line of a length n' in the same range with $J_{n'} = J_n$ on $[0, T+1]$, has a cell in state F at a relative time $\tau \le T$.

By Lemma 1 and Corollary 3 (with $N_0 = \max(15, T+3)$), every line of length $n \geq \max(15, T+3)$ has an F at a relative time $\tau \leq T < n-1$, that is, before time $2n-2$. The lines of lengths $2, \ldots, 14$ are synchronized by these rules (Proposition 5.2 of the note and its text). □

rule	T	t_0	p	lengths simulated	first F at relative time
δ_{14}	10	51	8	35 $(15, \ldots, 49)$	$7, \ldots, 10$
δ_{14}^{b}	14	66	3	39 $(15, \ldots, 53)$	$8, \ldots, 14$
δ_{14}^{c}	22	257	192	413 $(15, \ldots, 427)$	$7, \ldots, 22$
δ_{14}^{d}	12	95	48	116 $(15, \ldots, 130)$	$6, \ldots, 12$
δ_{14}^{e}	14	56	2	28 $(15, \ldots, 42)$	$9, \ldots, 14$

Table 1: Certificates for Proposition 5: the depth column of depths $0, \ldots, 2T+4$ repeats with period p from time t_0; one length per distinct signature (input cells at $\kappa \leq T+2$ and the return-chain values of C_∞, a slightly finer partition than needed) is simulated, together with every length $15 \leq n < T+3$. Program: `failall.py RULE T`.

The early-rows test Let g be a germ that determines C_∞ on the cells used below, and let the depth column of C_∞ for $D = 2T+1$ satisfy $V_{t_0} = V_{t_0+p}$. If a rule $\delta \supseteq g$ is a minimal-time solution, then for each of the finitely many lengths n of Corollary 3 no cell of R_n with $\tau \leq \min(T, n-2)$ is F, and $R_n(\tau, \tau) \neq C_\infty(n-1+\tau, n-\tau)$ for $\tau \leq \min(T, n-1)$ by Lemma 2.3 of the note. By Corollary 3, these conditions for the listed lengths imply the same conditions for every $n \geq N_0$: no further length adds a constraint on the early rows. Combined with the complete triangles of the lengths $2, \ldots, N_0 - 1$, they form a finite formula; if it has no model extending g, no minimal-time solution contains g. This test is the encoding `early:n:T` of `gfix.py`, with the lengths produced by `asymp.py`.

2 Uniform pre-firing configurations

Lemma 6. *Let $X \in W$, $N \geq 3$, and let δ be a minimal-time solution such that for every $3 \leq n \leq N$ the configuration of the line of length n at time $2n-3$ is X^n. Define δ' by $\delta'(x, y, z) := \mathsf{L}$ for every $(x, y, z) \in W^3 \setminus \{(X, X, X)\}$ with $\delta(x, y, z) = \mathsf{F}$, and $\delta' := \delta$ elsewhere. Then $\delta'(X, X, X) = \mathsf{F}$, no other neighbourhood without a border is mapped to F by δ', and δ' coincides with δ on every neighbourhood that occurs in the half-line or in a line of length $n \leq N$ before time $2n-2$. In particular, δ' satisfies every constraint that the lines of lengths $2, \ldots, N$ and the half-line impose on δ.*

Proof. The rule δ' satisfies the quiescence condition (1) of the note, since $\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \mathsf{L}$. The line of length 3 fires from XXX, so $\delta(X, X, X) = \mathsf{F} = \delta'(X, X, X)$. Let $(x, y, z) \in W^3 \setminus \{(X, X, X)\}$ with $\delta(x, y, z) = \mathsf{F}$. It does not occur in the half-line, which never fires (equation (2) of the note). If it occurred as the neighbourhood of a cell of the line of length $n \leq N$ at a time $t < 2n-2$, that cell would be F at time $t+1$; this is impossible for $t+1 < 2n-2$, and for $t = 2n-3$ the configuration is X^n, whose neighbourhoods without a border are all (X, X, X) (the line of length 2 has none). Changing the values of neighbourhoods that never occur changes neither the half-line nor these lines. □

So if the constraints of the lengths $2, \ldots, N$ (with the half-line encoded to any anti-diagonal), together with "the only neighbourhood without a border that gives F is (X, X, X)", are unsatisfiable,

then no minimal-time solution fires from X^n at every length $3 \le n \le N$. This is option `-Pint XXX` of `gcegar.py`; option `-upf X` adds the implied unit clauses "every cell at time $2n-3$ is X". The state $X = \mathsf{L}$ is impossible, because $\delta(\mathsf{L},\mathsf{L},\mathsf{L}) = \mathsf{L}$, and $X = \mathsf{B}$ is the case $X = \mathsf{A}$ after exchanging the auxiliary states. Mazoyer's solution fires from G^n, and so do the rules satisfying Balzer's condition (B3) in the strong reading.

3 A pumping lemma for the right band

The left-border lemma (Lemma 4.1 of the note) runs a band of chains along an anti-diagonal. The same pigeonhole argument applies to the cells next to the right border, read in the direction of time. It gives a necessary condition on the reflected triangle itself.

Lemma 7 (right band)**.** *Let δ have k states and synchronize the line of length n at time $2n-2$. Let $1 \le d \le \tau_0 \le n-2$ and $\ell \ge 1$, and suppose that the column $\tau \mapsto R_n(\tau, d)$ is ℓ-periodic on $[\tau_0, n-2]$. Then $n-1-\tau_0 \le (k-1)^d \ell$.*

Proof. For $d-1 \le \tau \le n-1$ put $\mathbf{b}(\tau) := (R_n(\tau,0), \dots, R_n(\tau,d-1))$. Since $R_n(\tau+1,\kappa) = \delta(R_n(\tau,\kappa+1), R_n(\tau,\kappa), R_n(\tau,\kappa-1))$ with $R_n(\tau,-1) = *$, there is a map Γ, depending on δ and d only, with $\mathbf{b}(\tau+1) = \Gamma(\mathbf{b}(\tau), R_n(\tau,d))$ for $d \le \tau \le n-2$; its arguments lie in $W^d \times W$ because no cell fires before time $2n-2$. At $\tau = n-1$ the band is F^d. Suppose $n-1-\tau_0 > (k-1)^d\ell$. The pairs $(\mathbf{b}(\tau), \tau \bmod \ell)$, $\tau_0 \le \tau \le n-2$, take at most $(k-1)^d\ell$ values, so there are $\tau_0 \le \tau_1 < \tau_2 \le n-2$ with $\mathbf{b}(\tau_1) = \mathbf{b}(\tau_2)$ and $\ell \mid h := \tau_2 - \tau_1$. For $0 \le s \le n-2-\tau_2$ the driving cells $R_n(\tau_1+s,d)$ and $R_n(\tau_2+s,d)$ lie in $[\tau_0, n-2]$ and differ by a multiple of the period, so they are equal, and induction on s gives $\mathbf{b}(\tau_1+s) = \mathbf{b}(\tau_2+s)$ for $0 \le s \le n-1-\tau_2$. For $s = n-1-\tau_2$ this reads $\mathbf{b}(n-1-h) = \mathbf{b}(n-1) = \mathsf{F}^d$, although $n-1-h < n-1$: a contradiction. □

Unlike the two barriers of the note, the hypothesis concerns cells of R_n, not of the half-line, so the lemma is no test on germs. The partial-triangle tests (cones of firing cells, `gfix.py`) imply it for every length they encode.

4 The architecture of Mazoyer's solution

The following facts about Mazoyer's six-state solution (file `results/mazoyer6.txt`, extracted from Duprat's Coq development) are established by simulation; they motivate the classes of rules tested in `LOG.md`.

Proposition 8 (computational)**.** *Let δ_M be Mazoyer's rule and let U be the set of neighbourhoods that occur in the lines of lengths $3, \dots, 120$ before their firing time.*

(i) $\delta_M(*,y,z) = \delta_M(\mathsf{L},y,z)$ *for every $(*,y,z) \in U$ (five neighbourhoods), and $\delta_M(x,y,*) = \delta_M(x,y,\mathsf{G})$ for every $(x,y,*) \in U$ (fifteen neighbourhoods); for every other state X at least eight of the fifteen right-border entries differ from $\delta_M(x,y,X)$.*

(ii) $\delta_M(\mathsf{L},\mathsf{L},x) = \mathsf{L}$ *for every working state x.*

(iii) Let $H(n)$ be the largest T such that every cell of R_n with $\kappa \le \tau - 4$ and $\tau \le T$ is L: up to the relative time $H(n)$, the reflected triangle is quiescent behind an echo of width four. Then $H(n) = 4, 4, 5, 6, 8, 9, 11, 14, 21, 31, 41, 201$ for $n = 8, 10, 12, 15, 20, 25, 30, 40, 60, 90, 120, 600$.

So the left border of Mazoyer's solution acts as a quiescent cell and the right border as a general, and the echo of the right end clears the line behind it until it meets the division point at about one third of the line from the right end ($H(n) \approx n/3$), where a new general starts a shorter line in the cleared cells. None of the five-state rules δ_{14}, $\delta_{14}^{\mathrm{b}}, \ldots, \delta_{14}^{\mathrm{e}}$ has either property: each uses border transitions that differ from the interior transition with the border replaced by any state, and behind their echoes the reflected triangles never become quiescent (Proposition 5 and `LOG.md`, §§20–21).

5 The complexity-15 half-line

The two germs of complexity 15 that survive the tests of the note are those of δ_{14}^{c} and δ_{14}^{d}. Let σ be the renaming $\mathsf{A} \mapsto \mathsf{G}$, $\mathsf{B} \mapsto \mathsf{A}$, $\mathsf{G} \mapsto \mathsf{B}$ of the states of δ_{14}^{d} (L and F fixed).

Lemma 9. *σ maps the thirteen transitions of the germ of δ_{14}^{d} that occur at times $t \geq 1$ onto the thirteen transitions of the germ of δ_{14}^{c} that occur at times $t \geq 1$. Let $n \geq 3$. In the names of δ_{14}^{d}, the line of length n is $\mathsf{A}\,\mathsf{B}\,\mathsf{L}^{n-2}$ at time 1 for both germs. Hence the constraints that the line of length n imposes on a completion of the germ of δ_{14}^{d} and, transported by σ^{-1}, on a completion of the germ of δ_{14}^{c} are the same. The only exceptions are the four entries $(*, \mathsf{G}, \mathsf{L})$, $(\mathsf{G}, \mathsf{L}, \mathsf{L})$, which the germ of δ_{14}^{d} fixes to A, B, and $(*, \mathsf{A}, \mathsf{L})$, $(\mathsf{A}, \mathsf{L}, \mathsf{L})$, which the transported germ of δ_{14}^{c} fixes to A, B.*

Proof. The germ of δ_{14}^{d} is $*\mathsf{GL}{>}\mathsf{A}$, $\mathsf{GLL}{>}\mathsf{B}$, $*\mathsf{AB}{>}\mathsf{A}$, $\mathsf{ABL}{>}\mathsf{B}$, $\mathsf{BLL}{>}\mathsf{B}$, $\mathsf{ABB}{>}\mathsf{B}$, $\mathsf{BBL}{>}\mathsf{L}$, $\mathsf{BLB}{>}\mathsf{B}$, $\mathsf{LBL}{>}\mathsf{A}$, $\mathsf{BBA}{>}\mathsf{L}$, $\mathsf{BAB}{>}\mathsf{B}$, $\mathsf{LBB}{>}\mathsf{L}$, $\mathsf{BBB}{>}\mathsf{B}$, $\mathsf{LBA}{>}\mathsf{B}$, $\mathsf{ABA}{>}\mathsf{A}$. That of δ_{14}^{c} is $*\mathsf{GL}{>}\mathsf{G}$, $\mathsf{GLL}{>}\mathsf{A}$, $*\mathsf{GA}{>}\mathsf{G}$, $\mathsf{GAL}{>}\mathsf{A}$, $\mathsf{ALL}{>}\mathsf{A}$, $\mathsf{GAA}{>}\mathsf{A}$, $\mathsf{AAL}{>}\mathsf{L}$, $\mathsf{ALA}{>}\mathsf{A}$, $\mathsf{LAL}{>}\mathsf{G}$, $\mathsf{AAG}{>}\mathsf{L}$, $\mathsf{AGA}{>}\mathsf{A}$, $\mathsf{LAA}{>}\mathsf{L}$, $\mathsf{AAA}{>}\mathsf{A}$, $\mathsf{LAG}{>}\mathsf{A}$, $\mathsf{GAG}{>}\mathsf{G}$. Applying σ to the last thirteen transitions of the first list gives the last thirteen of the second. In both germs the first two transitions are used only at $t = 0$ (the general is the only cell in state G of δ_{14}^{d}'s half-line, and $(*, \mathsf{G}, \mathsf{L})$, $(\mathsf{G}, \mathsf{L}, \mathsf{L})$ do not occur in δ_{14}^{c}'s half-line at $t \geq 1$, by the correspondence). At time 1 the line of length $n \geq 3$ is $(\delta(*, \mathsf{G}, \mathsf{L}), \delta(\mathsf{G}, \mathsf{L}, \mathsf{L}), \mathsf{L}, \ldots, \mathsf{L})$, which is $\mathsf{A}\,\mathsf{B}\,\mathsf{L}^{n-2}$ for δ_{14}^{d} and $\mathsf{G}\,\mathsf{A}\,\mathsf{L}^{n-2}$, that is $\sigma(\mathsf{A}\,\mathsf{B}\,\mathsf{L}^{n-2})$, for δ_{14}^{c}. From time 1 on, the line evolves by the table alone, so a completion δ of the first germ and the completion $\sigma \circ \delta \circ \sigma^{-1}$ of the second (after the four entries above are set as stated) produce corresponding lines of every length $n \geq 3$. The line of length 2 reads $(\mathsf{G}, \mathsf{L}, *)$ at time 0 in both germs, which corresponds to two different entries. □

Consequently the completion problems of the two labellings for the lengths $3, \ldots, N$, with the four entries left free, have a common relaxation; if it is unsatisfiable, both germs are refuted and every five-state minimal-time solution has $c_{78} \geq 16$.